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Literature Review

Zhongqi Xiu

2026/02/13

Article

Signatures of fractional charges via anyon–trions in twisted $MoTe_2$

https://doi.org/10.1038/s41586-026-10101-w

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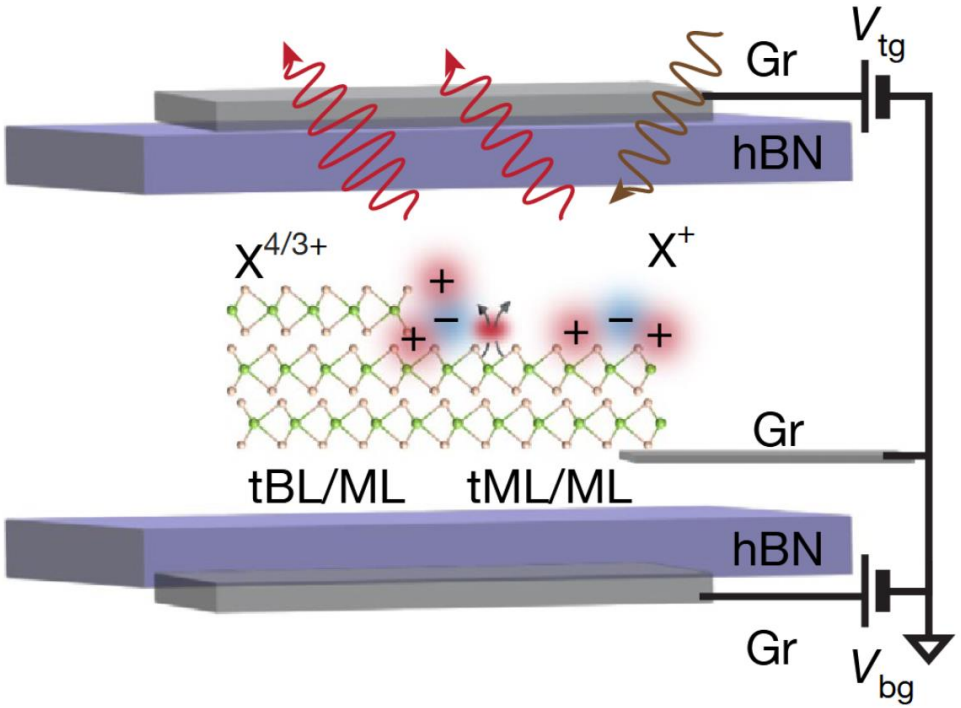
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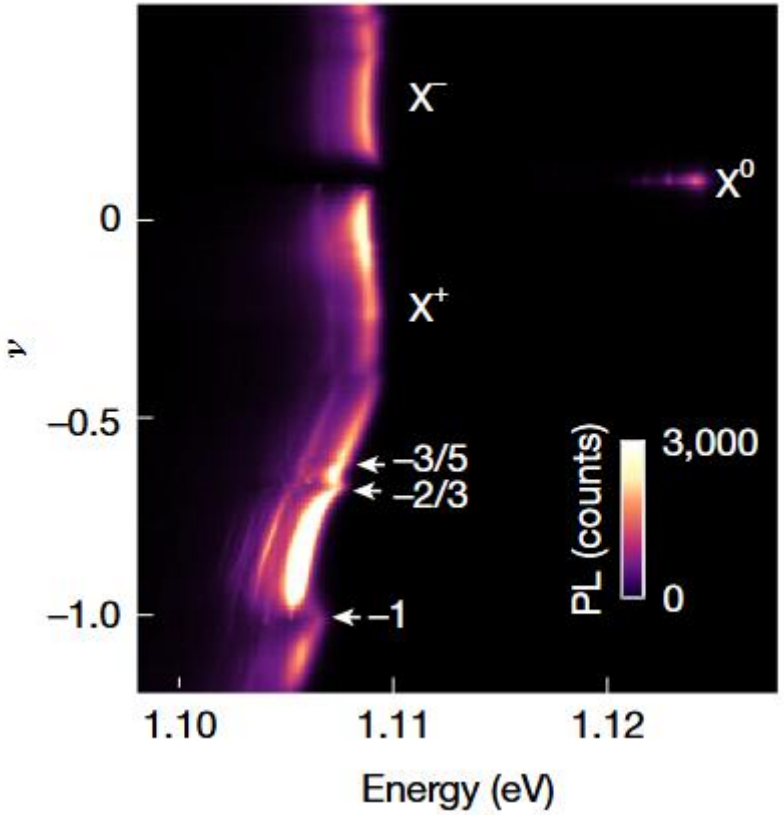
Weijie Li[1], Christiano Wang Beach[1], Chaowei Hu[1], Takashi Taniguchi[2], Kenji Watanabe[3], Jiun-Haw Chu[1], Ataç Imamoğlu[4], Ting Cao[5], Di Xiao[1,5] & Xiaodong Xu[1,5]✉

Device characterization

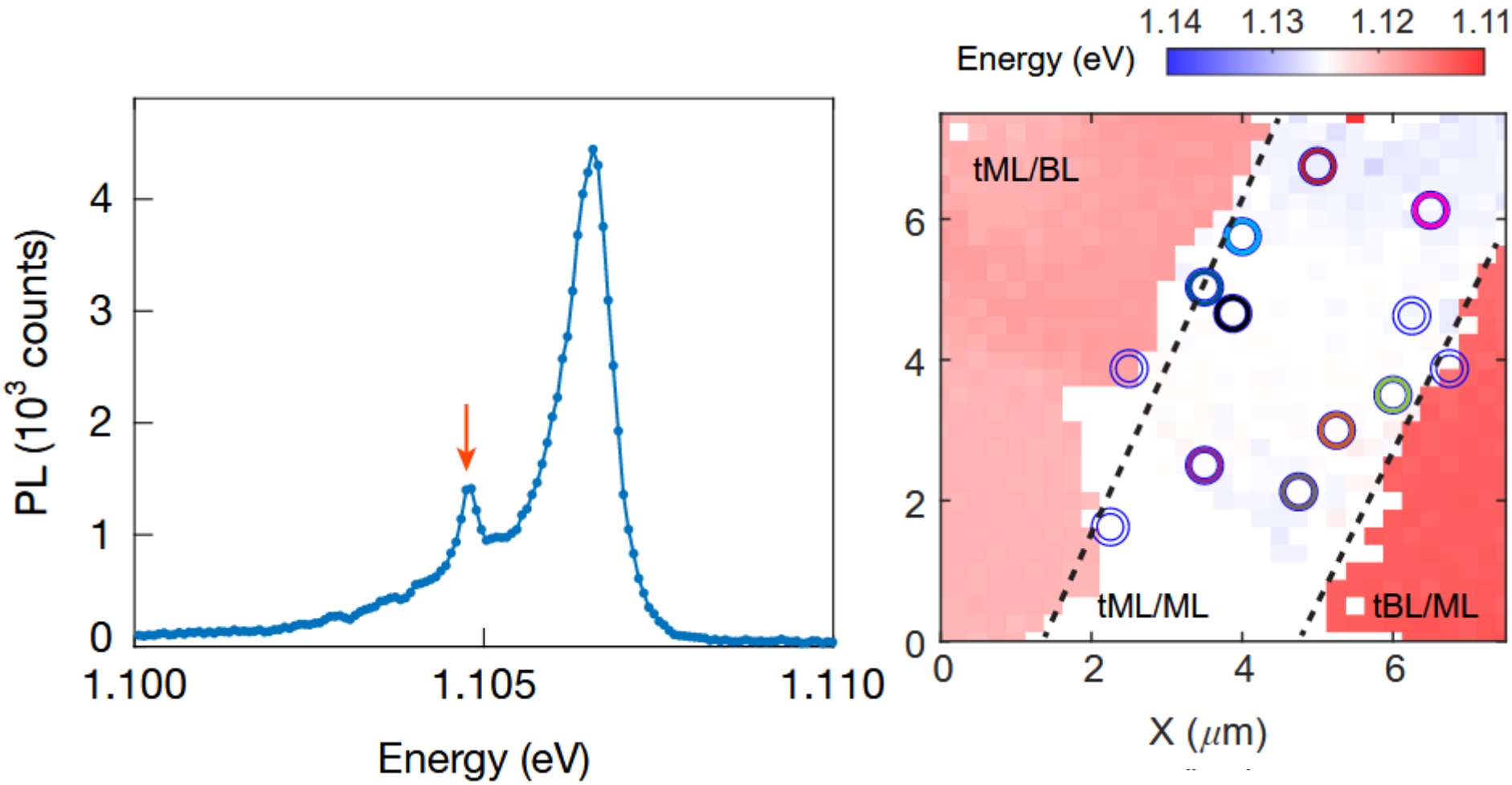
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T-monolayer/ monolayer with H-stacked bilayer



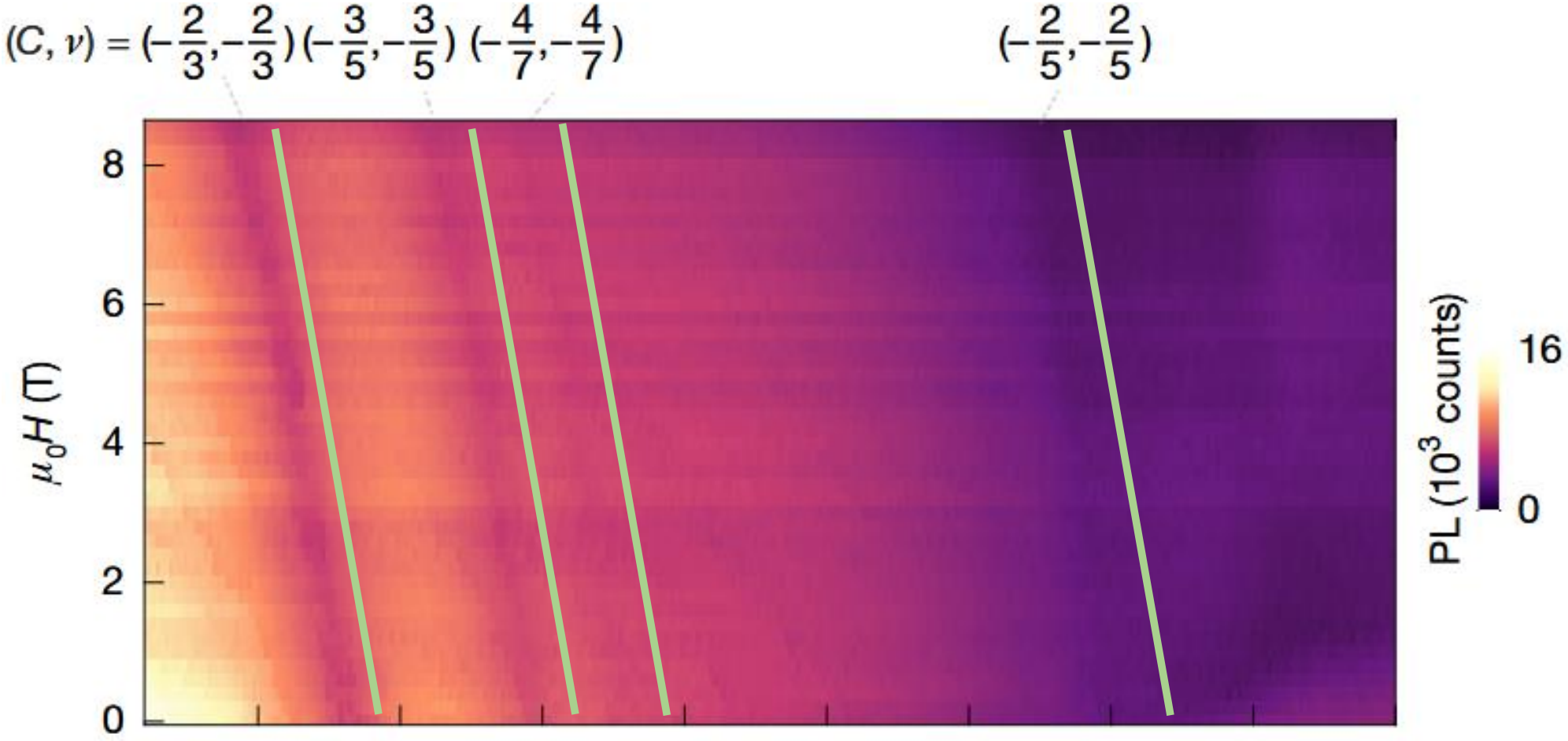
Identify new peak



New narrow PL peak around the trion signal (linewidths about 100 ueV)

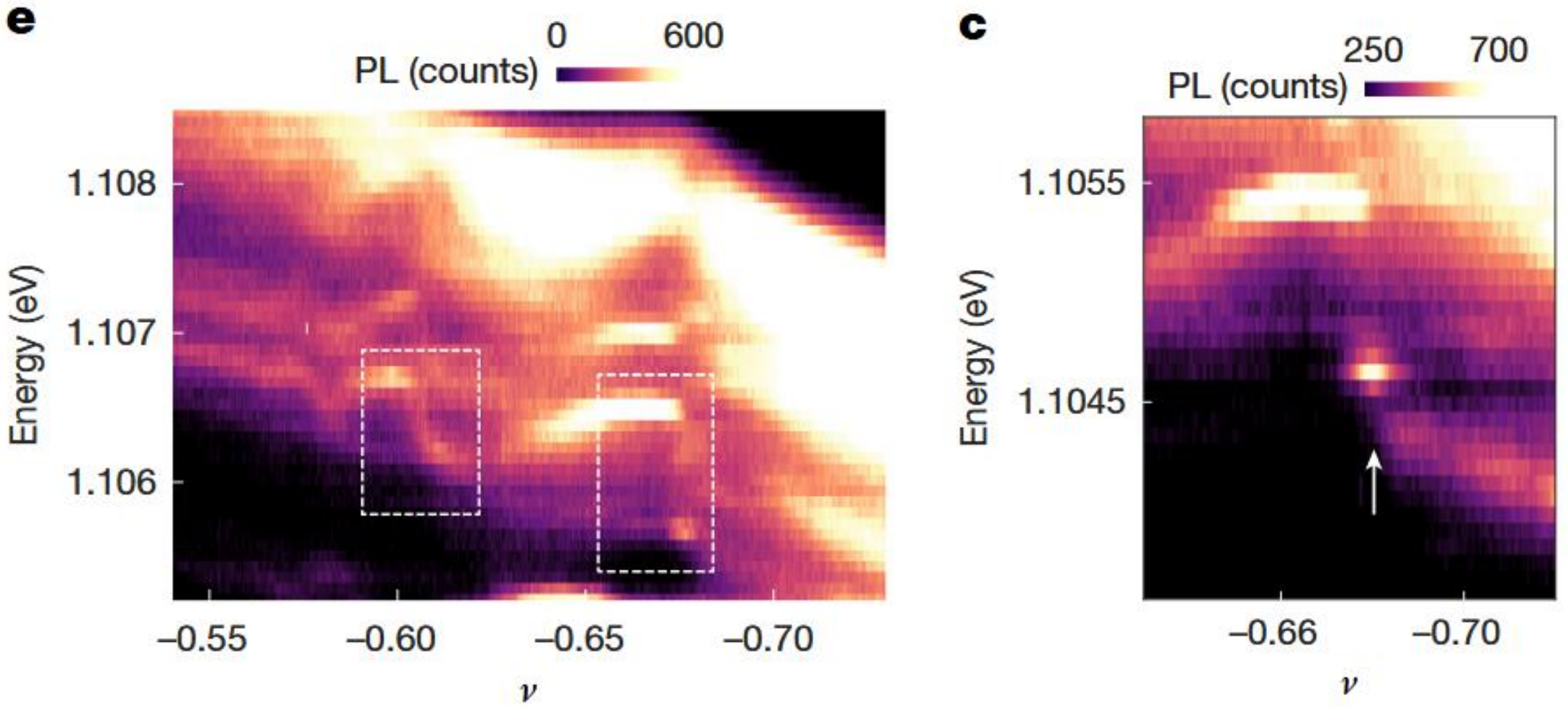
Fractional Chern insulator states

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-2/3 FCI state

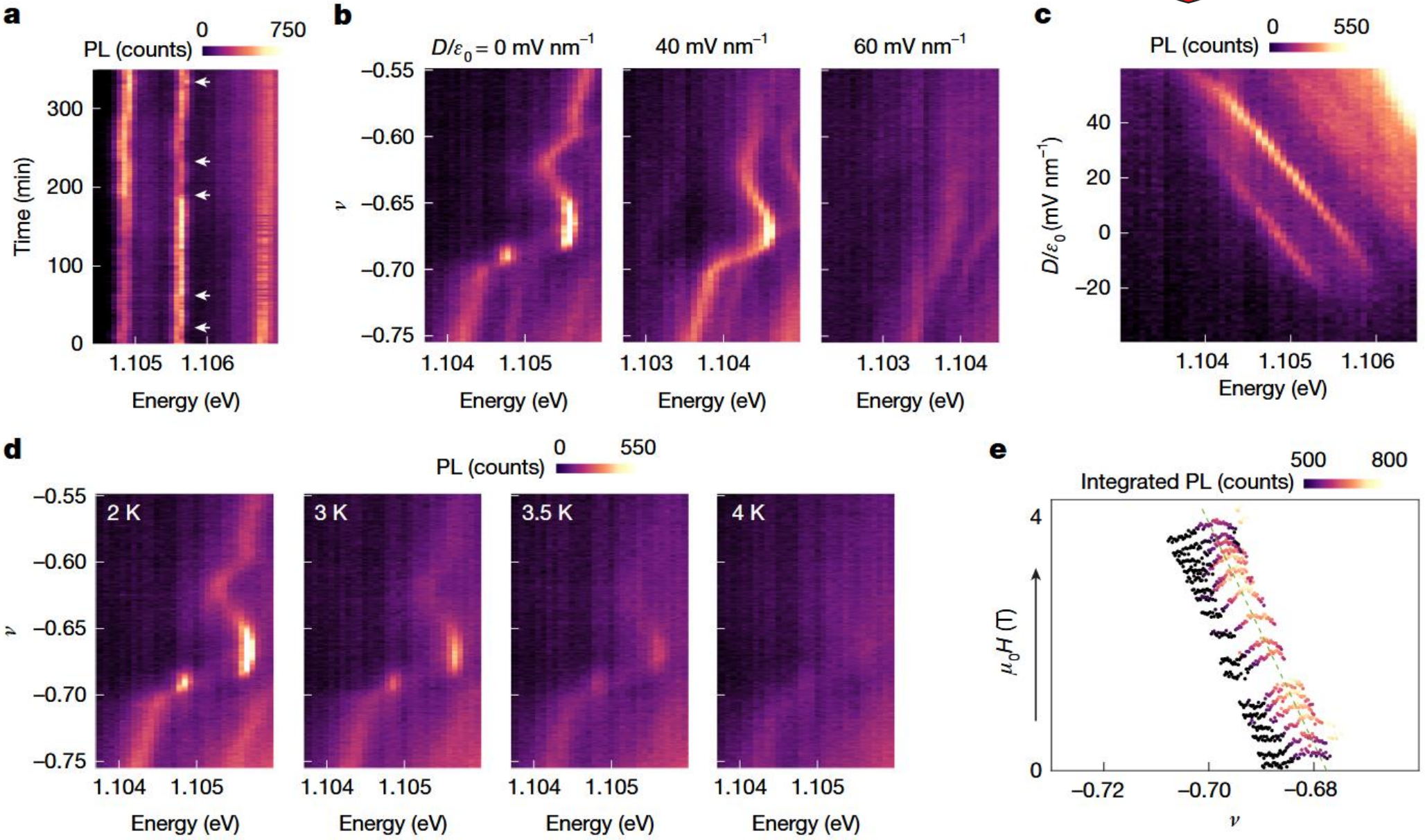
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Phase transition from FCI to the anomalous Hall metal phase

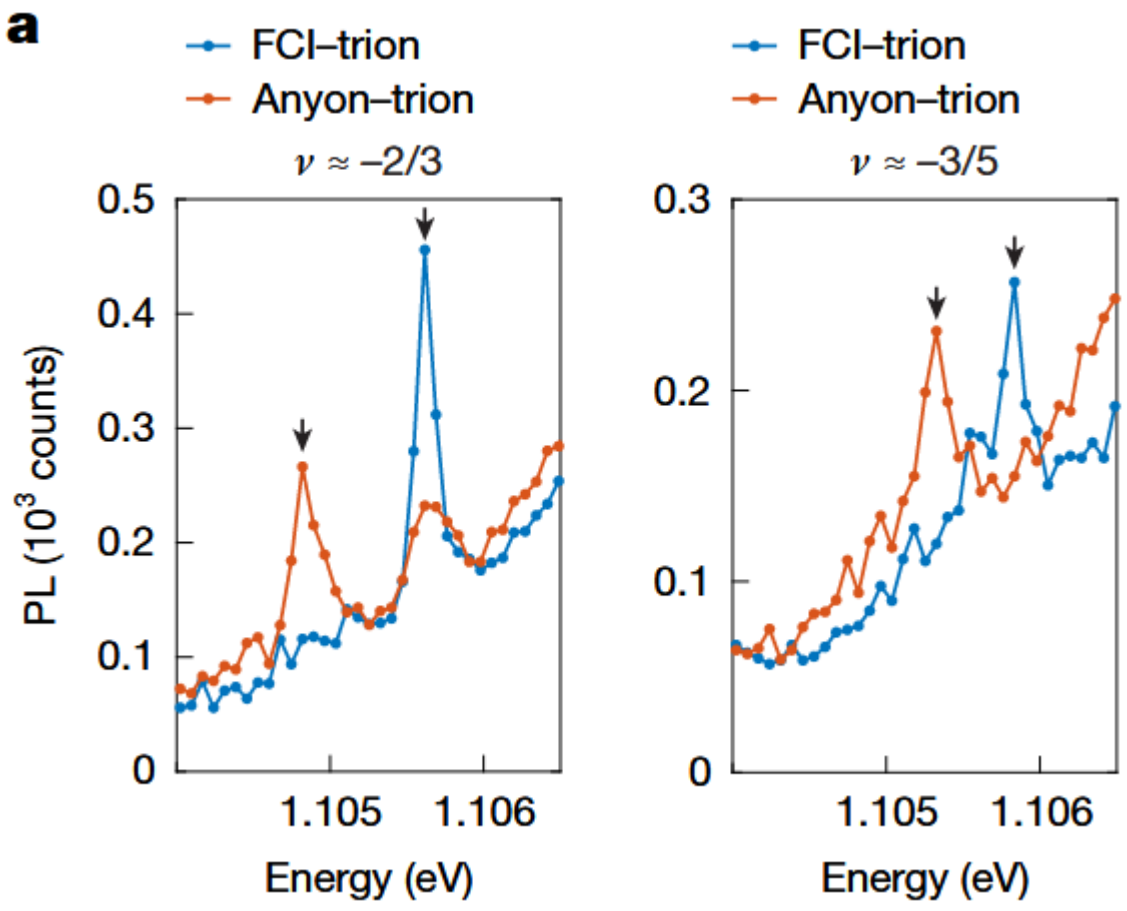
Evidence of anyon–trions

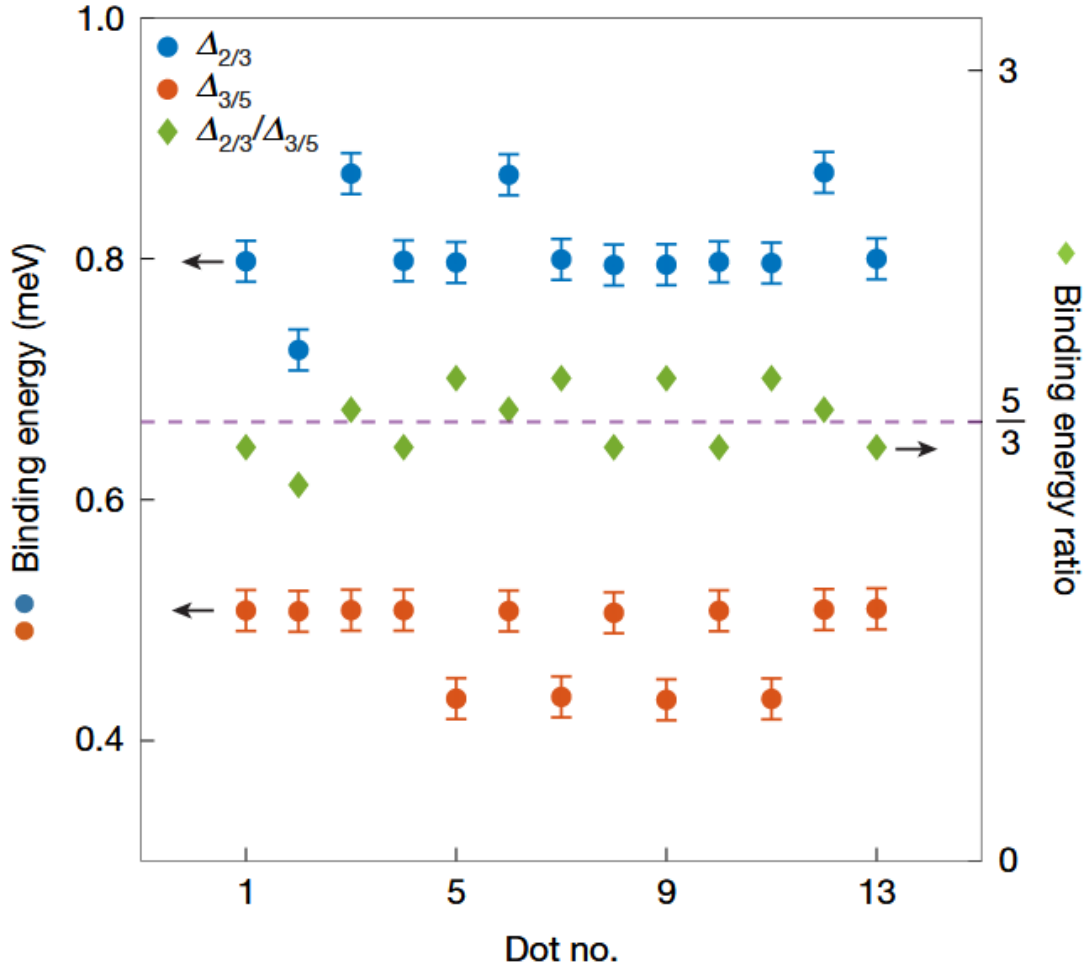
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Anyon–trion binding energy

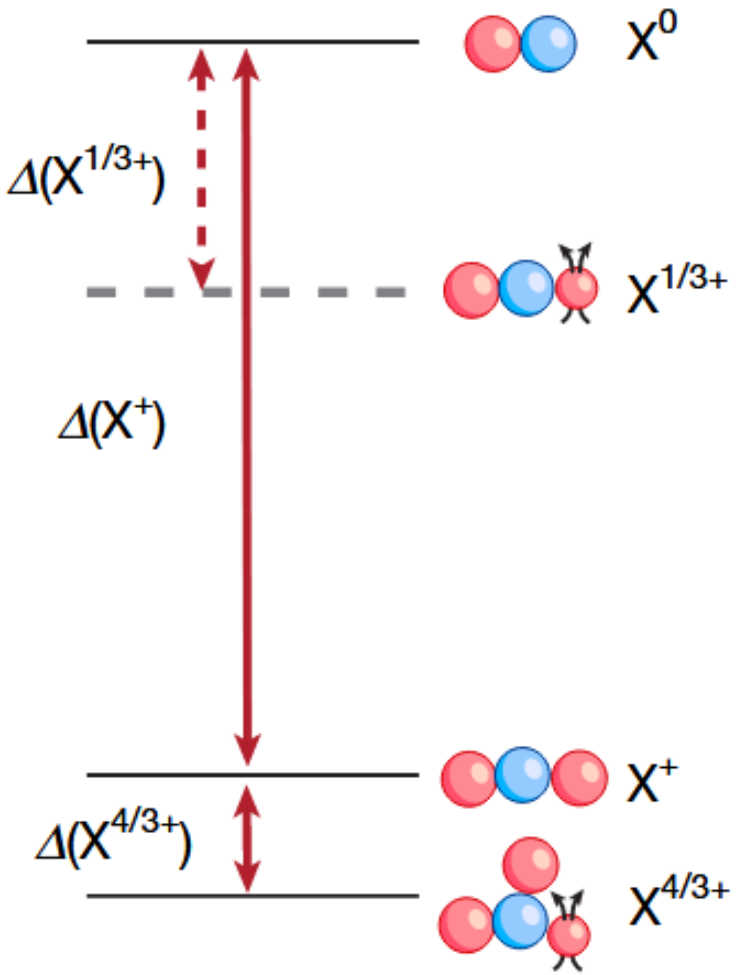
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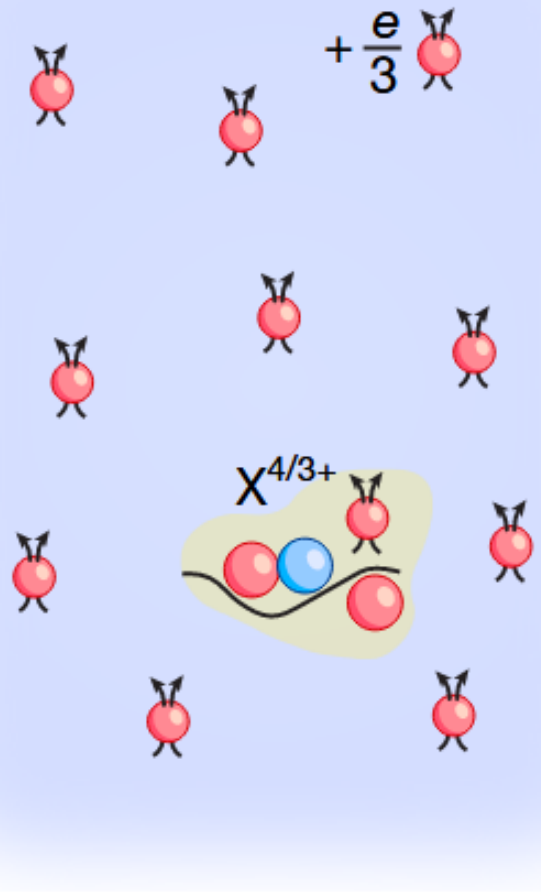


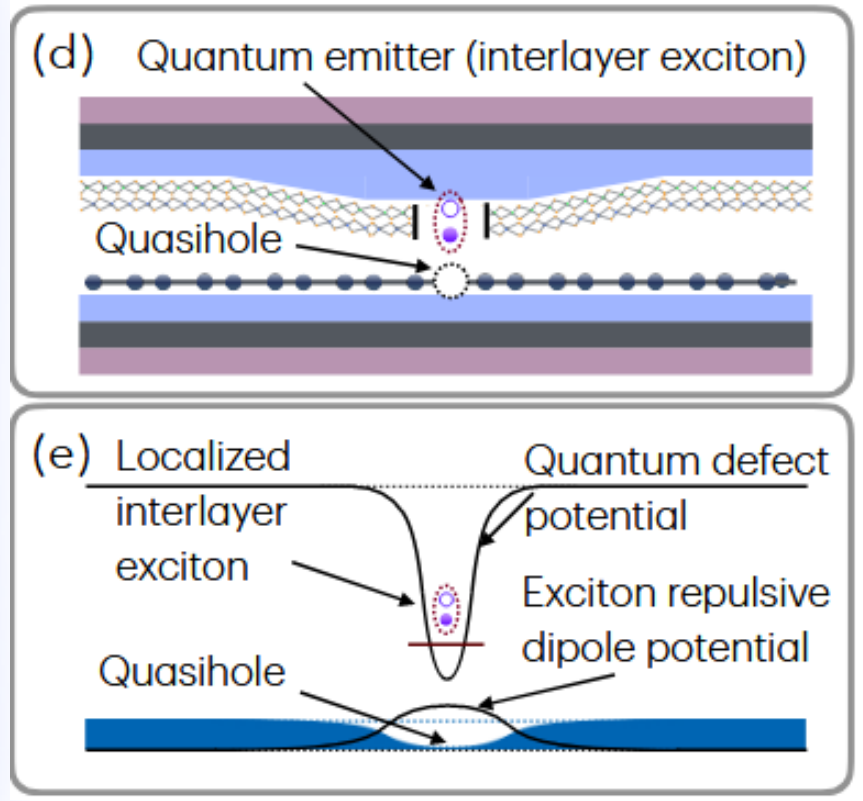


Why trion not exciton

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PRX Quantum 7, 010325



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THANKS

Speaker: Zhongqi Xiu